How-To Library

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Matching Multi-language Data in MDM SaaS

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Abstract

This article contains information about adopting a unified approach for multi-language matching in MDM SaaS. Using this approach, organizations can use a single match model instead of creating separate match models for matching multi-language data in MDM SaaS.

Supported Versions

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Introduction

In today's interconnected world where data flows seamlessly across borders and linguistic boundaries, businesses face the intricate challenge of accurately matching names and other entities across diverse languages and regions. From customer databases to identity verification systems, the ability to match names effectively is essential for maintaining data integrity, ensuring regulatory compliance, and facilitating seamless operations.

However, achieving accurate name matching across multiple languages presents a myriad of challenges. Variations in naming conventions, character encoding, and cultural nuances can confound traditional matching algorithms, leading to inaccuracies and inconsistencies in matching results.

In this article, we delve into the complexities of cross-language name matching, explore real-world scenarios, and examine the behavior of existing matching algorithms. Through examples, we explain how Informatica MDM SaaS can help organizations overcome their challenges when matching names across different languages.

Overview

Before getting into matching multi-language data, it's essential to understand how matching works in Informatica MDM SaaS and to familiarize with population files.

To run matching, a match model must be created, and it can contain the following objects:

- A set of declarative rules that can perform exact and fuzzy matches.
- A machine learning (ML) model that's trained using customer's data.

The following image outlines the primary steps involved in matching:



Of steps 2, 3, and 4, you can choose to use all three, or only the ones you want.

1. Candidate selection

Match candidates are record pairs that are possible matches. For each record, we select other records that are possible matches. These records will be analyzed further to determine if they are a match or not.

For any data model, you can choose the attributes to determine the possible matches. For example, you can choose Name, Address, IDs, and Phone numbers.

2. Exact match

Exact match is a deterministic match that matches records containing identical values. Configure the exact match strategy if the quality of data is good.

Use the exact match strategy for fields that contain unique data, such as IDs, postal codes, and industry codes.

3. Fuzzy match

Fuzzy match is a probabilistic match that includes variations in data patterns, such as spelling variations, misspellings, transpositions, omissions, and phonetic variations. It ensures that the match process matches records with data variations.

For example, use the fuzzy match strategy to match records based on fields that contain addresses and names of people or organizations. Use exact match and fuzzy match rules together for high-quality results with better performance.

4. AI match

AI match in MDM SaaS uses advanced machine learning (ML) models to enhance the accuracy and efficiency of matching process. You can train an ML model to predict whether record pairs are a match or not. AI match dynamically adjusts matching parameters and strategies during match training.

During the training process, label the presented record pairs as match or not a match. The model learns from the labeled record pairs to predict whether other record pairs are a match. When you train an AI match model using your data, the data remains fully secured in your tenant, and neither the data nor the trained model is shared in any way.

5. Survivorship and merge

Merge is the process of creating a master record from multiple matching source records. Survivorship is the mechanism used to determine the best data from the source records to use while creating the master record when a merge happens or updates come in from source systems.

The survivorship process begins after the match process identifies the matching source records. Survivorship configuration specifies which data from all the matching source records must survive in the merged master record. Survivorship configuration includes survivorship rules that are based on source ranks, source last updated date, block survivorship, and deduplication criteria.

Populations

A population is a ruleset that defines specific characteristics of data, such as names, addresses, and other identification details, in the records that you want to match.

The population ruleset accounts for variations and errors in name, address, and other data. MDM SaaS uses populations to perform fuzzy matching of records in different languages and different countries. We have 70 populations available out of the box.

For example, a population contains the following information:

- Common family names in each population. For example, Smith is common in the US, but not in China.
- Large number of uncommon names that tend to have the most error and variability.
- Common nicknames. For example, Bob is a nickname for Robert.
- Common abbreviations. For example, Inc. is an abbreviation for incorporated.

For a list of supported populations, see Supported Populations.

Match models for multi-language data

You can create a single match model for matching multi-language data in MDM SaaS instead of creating separate match models for different populations.

Creating a match model to support multiple populations

Rather than creating separate match models for each language or region, organizations can adopt a unified approach to multi-language matching and use a single match model. To ensure consistent results across diverse datasets and streamline match processes, you can use a centralized match model that uses different population files. These population files are tailored to specific linguistic and regional contexts.

When creating a match model, you can select a default population. For records in another language or from another region, you can override the default population by assigning another population for those specific records.

Note: If you need to have different match rules within a model for various countries, you can use "Multi-language matching by using segment matching" on page 6.

Two types of multi-language matching

You can match records with the following types of matching:

- Same-script matching. When the data is written in the same script.
- Cross-script matching. When the data is written in different scripts, such as the native script and the Romanized script.

Let's assume that you have data from two different countries, USA and Japan. You can then use the following population files:

- For data from USA, use the `USA` population.
- For data from Japan, in the Japanese script or transliterated and written in Roman script (English), use the `Japan` population.
- For data from Japan, which is only transliterated in the Roman script (English), use the `Japan_r` or `International` population.

If you have more data from USA than Japan, then set the default population for the model to `USA`, or else set it to `Japan`. To override the default population, specify another population for the records during ingress.

To determine which population to use for a given record, consider the following guidelines:

- Check the country code field in the address.
- Check the country code in the phone number.
- If the email is country-specific, check the domain within the email address.
- Utilize character-set recognition on the name.

Now you can use the following types of multi-script matching:

Same-script matching

If two records have data in the Japanese script, then they must use the Japan population for matching. Otherwise, they don't match.

Example 1:

- Default Population: `USA`
- Record 1 Name: 東京 (Population: `Japan`)
- Record 2 Name: 東京 (Population: `Japan`)

These records can match with exact or fuzzy declarative rules because both the records use the `Japan` population. The `Japan` population overrides the default population.

Example 2:

- Default Population: `Japan`
- Record 1 Name: 東京
- Record 2 Name: 東京

These records can match with exact or fuzzy declarative rules because both the records use the `Japan` population.

Example 3:

- Default Population: `USA`
- Record 1 Name: 東京
- Record 2 Name: 東京

These records can match with exact rules. They might also match with fuzzy rules because they are exactly the same. However, if there is even a slight variation, then they don't match because they use the `USA` population.

Cross-script matching

For data that's mixed and stored in multiple scripts, we can do cross-script matching. For example, one of the records is in the Japanese script and the other is in the Roman script. The cross-script matching works for some populations that support two scripts, such as Japanese, Chinese, and Arabic.

Note: Cross-script support refers to matching between native script and Roman script.

The following examples explain how cross-script matching works:

Example 1:

- Default Population: `USA`
- Record 1 Name: Tokyo (Population: `Japan`)
- Record 2 Name: 東京 (Population: `Japan`)

When you use cross-script matching, these records match based on fuzzy rules. They don't match with exact rules because both records rely on the `Japan` population file. The `Japan` population overrides the default population.

Example 2:

- Default Population: `USA`
- Record 1 Name: Tokyo
- Record 2 Name: 東京

These records can't match either with exact match or fuzzy as the `USA` population is used to match both records, and this population file can't read Record 2.

Example 3:

- Default Population: `USA`
- Record 1 Name: Tokyo (Population: `USA`)
- Record 2 Name: 東京(Population: `Japan`)

These records can't match either with exact match or fuzzy, as records use different population files, and they won't find each other.

Multi-language matching by using segment matching

In case you have country or region specific declarative rules, you can use segment matching to specify which country a particular match rule applies to. The field used to segment the data for segment matching can only be an exact match field.

For example, in a declarative rule, if you specify the segment value as Country = Japan, then that rule skips for records where the country isn't Japan.

Multi-language matching using the International population

For datasets primarily composed of data in English but spanning multiple countries, the International population can be used. This population encompasses a broad range of naming variations encountered across different regions, which use the Roman script. You can also use the International population when no specific population file is available for a given country.

Use the `Arabic_m` population for all Arabic speaking countries.

For Spanish-speaking South American countries for which country specific population aren't available, use any of the Spanish-based South American populations. For example, for Ecuador, you can use the `Colombia` population.

Conclusion

In navigating the complexities of cross-language name matching, Informatica MDM SaaS offers a robust solution. By leveraging language-specific population files, implementing fuzzy matching algorithms, and AI matching, Informatica empowers organizations to achieve reliable matching results.

As businesses expand globally, investing in MDM becomes crucial for maintaining data integrity and driving operational efficiency. With Informatica, organizations can navigate the multi-language maze with precision, ensuring impactful outcomes in a globalized economy.

For more information about matching using Informatica MDM SaaS, see Configure Match and Merge.

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